Energy Correlators as a probe of the hard process in Run 24 $p-p$ collisions at $\sqrt{s} = 200\ GeV$ with the sPHENIX detector at RHIC

by

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Abstract

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This dissertation consists of three parts in addition to a literature review establishing the state of the art of jet physics and the Energy-Energy correlator at high energies...

Part 1: Hardware This part discusses the physical hardware used to make the measurements of the energy in the jets created by the proton-proton collisions. This part contains a discussion of the sPHENIX detector and an in-depth look at the relevant susbsystems for the meausements in this analysis. In addition, the Monte Carlo methods and models to discern the expected respnse and calibrate the detector are discussed in detail, with both subparts coming together in a disucssion of backgrounds and error calculation in general and for the observables at hand.

Part 2: Technicalities This part builds from first principles up to the jet measurement in a theoretical light, and then discusses the additional computational techniques on display that will provide the bridge between the theory of these observables and the practical application to sPHENIX. This part continains a chapter that is a pared down discussion of a midstream paper proposed as part of the thesis process that establishes the safety of this observable against jet finding choices.

Part 3: Experimental Output This part is the meet and potatoes of the dissertation, discussing results from the experiment and the analysis in light of the previous parts and comparing to the world results from related systems.

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Chapter 1

Literature Review

1.1 Jets Definitions

The study of Quantum Chromodynamics in high energy collisions, such as those at the Relativistic Heavy Ion Collider or the Large Hadron Collider, is often carried out through investigation of the kinematics of jets. Jets, as objects, sit in the boundary between experiment and theory, being an experimental signature corresponding to final states of quarks and gluons produced in collisions. Jets are a cluster of final state particles that result from the showering and hadronization of the initial parton, that are identified in experiment through use of one of the multiple jet reconstruction algorithms.

1.1.1 Jet Reconstruction Algorithms

In Ryan Atkin's paper *Review of jet reconstruction algorithms* [1], Atkin provides an overview of the standard algorithms with a focus on practical implementation for experimental usage. Atkin discusses a number of algorithms that fall into two categories: Cone based and Sequential Clustering

Cone Methods

In Atkin's paper, Atkin discusses two iterative cone procedures–Iterative Cone with Progressive Removal (IC-PR) and with Split Merge (IC-SM)–and one more modern cone method, Seedless Infrared Safe Cone (SIS-Cone). These iterative cone procedures are described as simple to implement in an experimental context, however, they both suffer deficiencies from a theoretical perspective as they fail to be IRC safe [1] [2]. In principle, these clustering algorithms are rather similar, one takes the hardest particle in the final state, terms this a jet seed and draws a circle of a fixed size around it, summing the 4-momentum of all particles of within the circle. Then, the jet axis is compared to the jet seed, where a properly identified cone is defined as one where the condition in eq (1.1) is fulfilled.

$$\frac{p^{\mu}_{\text{jet}}}{|p_{\text{jet}}|} = \frac{p^{\mu}_{\text{seed}}}{|p_{\text{seed}}|} \tag{1.1}$$

If this condition is not fulfilled, one then starts again, setting the seed at the axis of the previously identified cone and draws a new circle of the same radius centered there. Once the jet is identified, the progressive removal procedure then removes all particles identified with this method and continues until there are no particles with a $p_T > p_T^{min}$ which is set by the user. In the split-merge procedure, one first identifies all seed particles, and builds the stable cones, but these are then treated as protojets. These protojets are then filtered by an additional $p_T^{\text{jet}} > p_T^{cut}$ threshold, then overlapping protojets are split or merged in descending p_T order depending on a merging parameter, f 0.5. If the sum of p_T for the shared particles is less than the fp_T^{sl} where p_T^{sl} is the lower p_T protojet, then the overlapping particles are split based on distance from the protojet axes and the protojets are updated. Else wise, the two protojets are merged into a single jet.

Both of these methods are IRC unsafe to some degree, but are useful in situations where quick jet identification is needed, e.g. CMS Cone and Pythia Cone are IC-PR and ATLAS Cone and JetClu midpoint cones are IC-SM . The IC-PR is collinear unsafe, as the progressive removal procedure inherently ignores any overlapping jets. The IC-SM however is infrared-unsafe as it introduces further parameters to filter pile-up and thus removes low energy contributions [2].

Sequential Clustering algorithms

The sequential clustering algorithms on the other hand are all IRC safe, working from the underlying assumption that all of the final state particles in a jet would have originated from a single initial parton, thus in theory, one can reconstruct that collection [3]. Three of these methods, anti-k_T, k_T and Cambridge/Achen, utilize the same mechanism with modifications to their metric. These methods, here referred to as k_T-type, rely on the evaluation of a distance metric of the form,

$$d_{ij} = \min\left\{p_{T,i}^{2n}, p_{T,j}^{2n}\right\} \frac{\Delta R_{ij}}{R} \qquad d_i = p_{T,i}^{2n} \tag{1.2}$$

Where $n = 1$ is k_T, $n = 0$ is the Cambridge/Achen and $n = -1$ is the anti-k_T, ΔR_{ij} is the angular distance between the jets in $\phi - y(\eta)$ space and R is the size of the jet, which functions as an estimated value of the average jet radius, a parameter set by the analyzer. One clusters particles together by finding $\min\left\{\{dij\}, \{d_i\}\right\}$ and, if the minimum value is one of these d_{ij} then one clusters those two, adds their 4 momentum together and recalculates the distance metric. If the minimum is a d_i–called beam distance, then the protojet is promoted to a jet and then removed from the set of candidates. While these methods are IRC safe [3], they do suffer from a similar tuning problem as the cone methods, where misidentification of jets is possible depending on the R value and p_T cutoff, however, both of these parameters can be estimated well and have led to a standard prescription of the anti-k_T method at a variety of R, usually prioritizing the $R = 0.4$ jet as the best balance between exclusion and false positive jet particles. Of note for the analysis of this system, the k_T-type algorithms can produce different collections of jet particles and different shapes of the final state of the jet. The anti-k_T method produces circular cones as a function of its metric, thus making it more directly comparable to the previous cone methods, while the k_T mehtod will produce final states of various sizes. This discrepancy proves useful in analysis of jet substructure observables, such as the primary family of observables, the Energy-Energy Correlators, in this thesis, as it allows for a lund Plane analysis of the jet structure that combines anti-k_T and k_T techniques to improve veracity of the analysis method

1.1.2 Quark and Gluon Jets

Standard Definitions: Event level label

Experimentally, there is significant interest in being able to discriminate quark and gluon jets in searches for Beyond Standard Model physics, searches for Dark Matter, in jet substructure analysis and in QGP physics. [4][5][6][].

Alternate Definition: Sample Populations

In their paper *An Operational Definition of Quark and Gluon Jets* [7] Kimiske, Metodiev and Thaler present a theoretical definition of jet flavor with an eye towards application at colliders. They aim to present a precise and practical definition of jet flavor at the hadron level, thus allowing for direct measurement from collider experiments rather than a measurement by proxy. While the gluon or quark jet is well defined at the level of the hard scattering, corresponding to an initiating parton of the primary vertex as was discussed above in sect. 1.1.2 [8] [9], this definition has limited realizability in experimental searches, and relies on an event lableling that is not supported by Quantum Field Theory approaches [7]. The determination of relative populations of quark and gluon jets in a dijet sample is one of the primary tasks of this thesis and will be discussed at length in chapters 11 and 12.

Kimiske et al.–from here referred to as KMT–set their work in the context of discerning the quark-gluon jet populations in samples rather than applying an event-by-event labeling scheme. KMT's work begins with the consdieration of two samples of data with jets at a fixed p_T range with the condition that one be gluon enriched and one quark enriched. In the specific test case presented in KMT, a Z-jet sample (gluon enriched) and a γ-jet sample (quark-enriched) are used.

1.2 Jet Substructure Measurements

1.3 N-Point Energy Correlator

Part I

Hardware: The Detector and Simulations

Chapter 2

The sPHENIX detector

The sPHENIX detector is part of the Relativistic Heavy Ion Collider (RHIC) complex at Brookhaven National Laboratories (BNL) in Upton, New York. RHIC is one of two major circular particle colliders in the world, the other of which is the Large Hadron Collider at CERN in Geneva, Switzerland. Over the course of the sPHENIX experiment which will run from run 23 until RHIC ceases operations after run 25, RHIC will be colliding gold nuclei and protons with run 23 having been dedicated to commissioning Au+Au running, run 24 dedicated to p+p with three weeks of Au+Au for further tracking commissioning and run 25 dedicated to physics running in Au+Au collisions. RHIC runs with a center of mass energy of $\sqrt{s} = 200GeV$ for both protons and gold, and additionally is the sole collider that collides polarized protons, which allows for more in depth studies of spin physics. At RHIC, sPHENIX's forerunner, the Pioneering High Energy Nuclear Interaction eXperiment (PHENIX), performed the first measurement of the Quark Gluon Plasma droplets [10] in nuclei-nuclei collisions.

sPHENIX is designed to offer significant upgrades to PHENIX, specifically providing increased coverage with hadronic and electromagnetic calorimeters, increasing effectiveness in studies of jets and heavy flavor at mid-to-high Bjorken x [11][12][13] Broadly, sPHENIX can

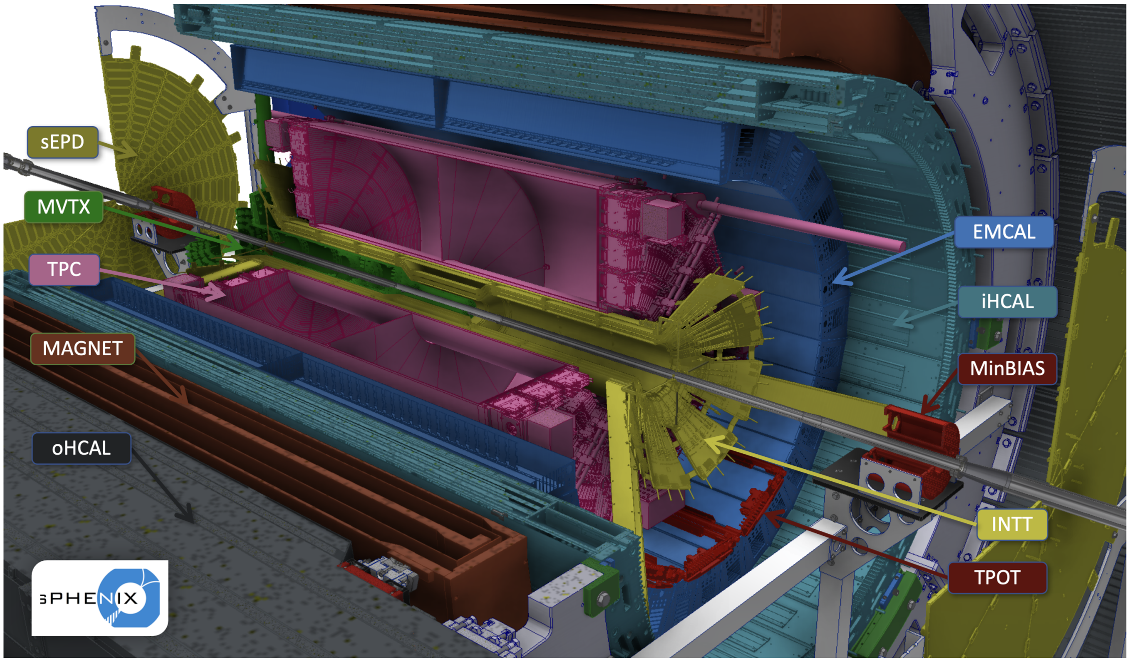


Figure 2.1: The sPHENIX detector. The Zero Degree Calorimeter is not picutred but is located further along the beam pipe.

be broken into three categories of subsystem: Calorimeters, Tracking and Event characterization.

sPHENIX is constructed around a 1.5 Tesla superconducting magnet, cooled by liquid helium, which was previously used on the BABAR experiment at SLAC, with the outer hadronic calorimeter sitting on the outer radius of the magnet, as see in fig 2.1

2.1 Calorimetery

2.1.1 Analog Digital Converters and Data Acquisition

Beyond the purposes and hardware of the detector itself, one can further characterize the subsystems of sPHENIX into streaming readout and non-streaming detectors, broadly distinguishing the tracking detectors from other subsystems. The non-streaming detectors all use a common data acquisition pipeline and data format. These detectors are readout to analog digital converters boards with 64 channels each, which are then grouped into sets of 3 called an ”XMIT” group, which is then read to the data format as a ”packet” of 192 channels. Each packet is transmitted from the detector to the computing center on a seperate fiber optic cable that is then read into a digitizer which forms groups of (at most) eight packets that are then assigned to individual data acquisition servers. The digitizing process allows for readback into the detector through a global trigger and timing module, where each DAQ server, can issue comands and readback to an event building logic that will then flow back to the detector controling the same XMIT groups. Each set of (at most) 4 XMIT groups is arranged into a crate with a single crate controller that recieves trigger and timing information distributed via a global timing and level 1 trigger module (GL1/GTM) that is then processed through a clockmaster for each digitizer rack, which will hold between 1 and 4 crates, depending of the needs of the subsystem. The data read out to each DAQ server is then stored in the ”Purschke Raq Data Format” or prdf file that provides an event structure consisiting of a header with some identifying information and seperate headers for each packet, but is otherwise comprehensible as an ASCII file. This event structure allows for easy offline combination of events across the calorimeters while keeping individual file size buffers relatively small, as each packet has timing information along with an event number relative to the run that is inhreited fom the timing module. This DAQ configuration is currently used for the calorimeters, the event plane detector and the minimuum bias detector, which allows for faster event building to verify data quality and alignment. The tracking detectors untilize a streaming readout format that creates event pools that are then associated with beam crossing timings provided by the RHIC clock and can be matched onto the GTM timing in offline analysis. This process is slow, and the fully implementation of event combination is still under developement, not accounting for event reconstructions which involves track matching across subsystems and will be discussed in section 2.2.5.

ADC Zero Suppression

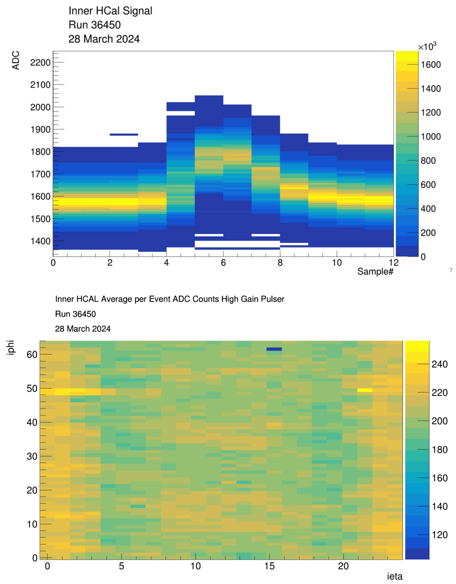
While this system allows for a simplified readout of the calorimetery system, one expects that, for the average event, most of the towers will be dominated by noise, thus one wastes signifigant computational resources on reading and storing data that will be immediately discarded. This decreases readout rates, runs the risk of filling the storage systems and increases the likelyhood of register errors that require restarting runs; in summary, these null data points give no useful physics and hurt our ability to consitently take data. The ADC and readout electronic systems are thus equipped with a zero supression system that alows for onboard discrimination and data reduction. At the FPGA level, each channel is individually evaluated by taking a post-pre measurement and comparing to a threshold derived from studies of average noise in each channel

$$ADC_{\text{sample 6}} - ADC_{\text{sample 0}} > E_{\text{cut off}} \tag{2.1}$$

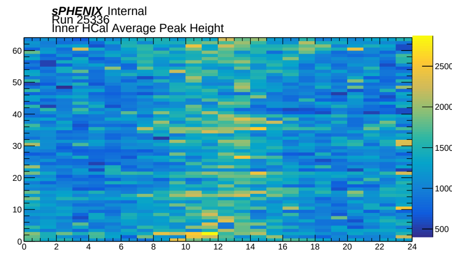
Where the evaluation shown in eq (2.1) assumes that the ADC readout has been properly timed in relative to the trigger such that the peak of well formed data should be centered near sample 6 (of 12, 16 or 31), and one can take a cutoff with resolution as wide as a single number and as granular as a tower-by-tower volume. Studies have been performed at sPHENIX to establish good thresholds for this zero-supression approach and validate that this will not introduce biasing into the detector [14].

2.1.2 Hadronic Calorimeters

sPHENIX's power in jet physics comes in large part from its calorimeter systems, including two seperate hadronic calorimeters. The outer and inner hadronic calorimeter (OHCal and IHCal respectively) are both divided into 24 bins in η with coverage of $|\eta| \leq 1.1$ and full φ coverage with 64 bins in φ, grouped in φ into 32 sectors in each calorimeter. Each



(a) Charge Injection Pulse. This system allows for direct testing of electronics without detector response



(b) LED pulse. This allows for testing of the full readback system and light response

Figure 2.2: Top: Presistence Waveform. Bottom: Energy distribution averaged across 10^5 events

hadronic calorimeter therefore has tower size of $\delta\eta = 0.092$ and $\delta\varphi = 0.098$. The outer hadronic calorimeter is constructed of steel and sits outside of the magnet, with inner radius of $r = 1.82$ m and outer radius of $r = 2.69$ m. The inner hadronic calormeter is constructed of aluminum and sits just inside the magnet, with inner radus of $r = 1/16$ m and outer radius of $r = 1.37$ m. Each tower corresponds to a readout board that sums readout from four scintillator paddles embeded into the calorimeter as show in fig. **??**. These interface boards form a single readout channel, and can be indvidually teste via a pulser system that injects charge into the interface board testing readback independent of scintillator response, and LED system that injects a fixed pulse of light into the scintilators to test behavior of the Silicon Photomultipliers (SiPMs). Output of each of these systems is shown in fig 2.2

Calibration

The hadronic calorimeters were intially calibrated through use of cosmic ray muons, matching the spectra to Monte Carlo generated through EcoMug and GEANT4 [**HCal˙Calib**]. This calibration is updated throughout the run via continued cosmic ray studies in between physics data taking, in addition to ongoing work on in-situo calibrations using tower-slope methods [**tower˙slope˙hcal**]. These calibrations have yeilded an average conversion factor for the OHCal of and the IHCal of

2.1.3 Electromagnetic Calorimeter

Moving in one layer from the hadronic calorimeter, the Electromagnetic Calorimeter (EMCal) has the same coverage in $\eta - \varphi$ space, but has 16 times as many towers with 96 bins in η and 256 bins in φ. The EMCal is constructed of blocks of tungsten with embedded scintilationg fibers, and had inner radius $r = 0.9$ m and outer radius of $r = 1.16$ m. Similarly to the HCals, the EMCal is also equipped with a pulser and LED, although the increased number of towers and higher varaiblility in response requires that different pulse widths be used for seperate sets of towers to prevent saturation and clipping on the LED pulse.

Calibration

The EMCal was calibrated to the π^0 mass through the $\pi^0 \to \gamma\gamma$ decay channel with additional corrections applied via the tower slope. The mean m_{π^0} and width are used as quality assurance plots to monitor radiation damage to the EMCal on an ongoing basis.

2.1.4 Zero Degree Calorimeter

The Zero Degree Calorimeter (ZDC) is a small transverse energy hadron detector that positioned outside of the experiment hall, in the tunnels along the beam pipe, 18 m from the interaction point on both side. The ZDC is used to detect spectator neutrons to allow for the calculation of multiplicity and event geometry, making it an overlapping detector between the calorimeters and the event characterization [15]. In addition, the ZDC provides a good proxy for total collision rates, useful for both real time monitoring of beam conditions and The sPHENIX ZDC is one of two detectors that was preserved from PHENIX, with minor repairs and upgrades to improve running. In pp collisions, the ZDC provides additional utility as a spin momentum detector that allows for measurment of the spin patterns of the RHIC beam through the Vernier Scan. This functionality is pivotal for the development of sPHENIX's spin physics program in pp and its extensions into the continuing physics goals at the Electron Ion Collider (EIC)[15]. For the purposes of this work, the ZDC is not utilized for either spin phyiscs or trigger capabilities and serves primiarily to help establish cross-sections for error determination. While this work does not dive into the subject of spin depenendence, such topics are a natural extension and present an intresting avenue for future studies in this vien to be done with the sPHENIX Run 24 data set.

2.2 Tracking

The sPHENIX tracking system consits of the Silicon tracking and vertexing detectors and the gas based trackers. These systems are designed for fast readout, precise momentum resolution and secondary vertex detection

2.2.1 TPC Outer Tracker

The TPC Outer Tracker (TPOT) was designed as a minimal add-on to improve tracking effeciency and resolution by providing additional track points.

2.2.2 Time Projection Chamber

The Time Projection Chamber (TPC) is a gas based detector designed to track

2.2.3 Intermediate Silicon Tracking Detector

The INTermediate Silicon Tracking detector (INTT) consists of two layers of semiconducting silicon

2.2.4 Micromegas Vertex Detector

The Micromegas Vertex detector (MVTX) is a silicon based vertexing detector based on a similar system at the ALICE experiment of the Large Hadron Collider (LHC). This detector sits closest into the beam pipe and is instrumental in the open heavy flavour program of sPHENIX [11] The MVTX is comprised of three layers ofof staves in concentric rings. Each of these is then further divided into

2.2.5 Event Reconstruction and Track Matching

2.3 Event Characterization

2.3.1 sPhenix Event Plane Detector

2.3.2 Minimum Bias Detector

The sPHENIX Minimum Bias Detector (MBD) is the other detector preserved from PHENIX, when it was called the Beam-Beam Counter (BBC). This detector is situated

2.4 Triggers

The sPHENIX detector has a number of triggers at its disposal to allow for online event type selections for the triggered readout system: the EMCal, the HCals, the ZDC, the MBD, the sEPD and–for run 2024–the TPC. Starting in run 2024, available triggers are ZDC and Minimum Bias in single and coincidence modes, a restricted vertex Minimum Bias trigger, four jet triggers at differing minimum energy thresholds (6-12 GeV with variation through the run), four photon triggers at differing minimum energy thresholds (2-10 GeV with variation through out the run), random triggers, clock triggers and jet with Minimum Bias triggers. In order to decrease busy signals, the overall trigger rate is scaled down to 8 kHz, taking 100 Hz of clock triggers as a control, full rate rare event (jet and photon) and minmum bias filling the remaining.

2.4.1 Jet Triggers

2.4.2 Photon Triggers

2.4.3 Minimum Bias Trigger

Chapter 3

Monte Carlo Simulations

For proton-proton collisions, sPHENIX has 10^8 fully reconstructed events from Pythia 8 and Herwig 7 for 2→2 and minimum bias (soft QCD) events that are then reconstructed using GEANT. While there is a jet sample that has been performed with the detroit tune [16] and there is an existing, equivilent tune for Herwig [17], in this work, I focus on the default tuned sample as there is a wider dataset from which to draw. data from the 2 and three point energy correlator, and its important for training of our discriminator

Chapter 4

Determining Backgrounds and Errors

Part II

Technicalities: The finer points of theory and computing

Chapter 5

The Energy Correlator and the primary vertex

Considering the limitations of Calorimetry only measuremet with RHIC one is So the energy correlator alows a look back, I guess thats the relevant bit? Point being, if the jet points back to the primary vertex, the share of the energy between the jets and the composition of those jets is directly determined by the $2 \rightarrow 2$ scattering and is thus sensitive to the parton distribution functions of the beam particles. Starting from the definition of the energy correlator given in [18]:

$$\mathcal{E}_N = \int \frac{\mathrm{d}\Omega}{2} \int \mathrm{d}\vec{n}_{12} \frac{\langle \varepsilon\left(\vec{n}_1\right) ... \varepsilon\left(\vec{n}_2\right) \rangle}{Q^2} \delta\left(\vec{n}_1 \cdot \vec{n}_2 - 2|\vec{n}_1|||\vec{n}_2| \cos\theta\right) \tag{5.1}$$

Where $\varepsilon\left(\vec{n}_i\right)$ is the energy flow operator defined as

$$\varepsilon\left(\vec{n}_i\right) = \lim_{r\rightarrow\infty} r^2 \int_{-\infty}^{\infty} \mathrm{d}t \vec{n}^i T_i^0 \left(t, r\vec{n}\right) \tag{5.2}$$

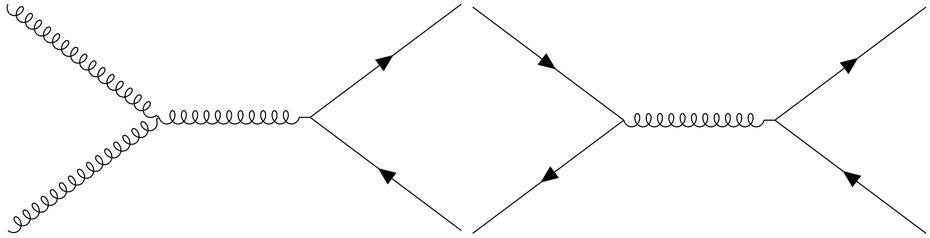


Figure 5.1: $p + p \to q\bar{q} + X$ events, these dijet-types are possible to be produced from gluon or quark interchange, probing the square of the gluon or quark distributions

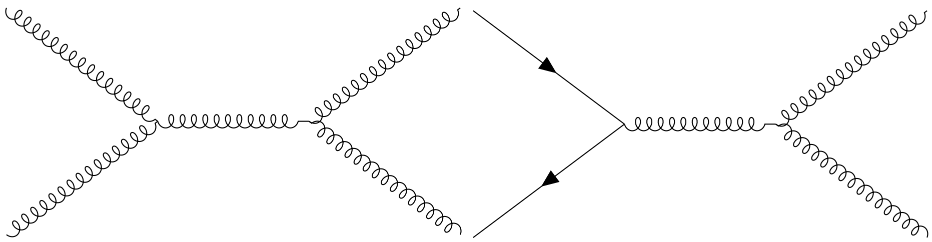


Figure 5.2: $p + p \to gg + X$ events, these dijet-types are possible to be produced from gluon or quark interchange, probing the square of the gluon or quark distributions

5.1 The primary vertex and interaction

The goal of this work is to extract gluon distributions from the jet, thus one must consider those intial verticies that (to first order) produce the gluon-quark, gluon-gluon, and quark-quark dijet events. To this end, one considers the following tree level diagrams While the processes are not imediately apparent in the final state of the jet, one notes that the two and three point energy correlators give a useful distinction as the jet function can be divided out, leaving a vertex dependent term at $R_L \approx 2R_{\text{jet}}$

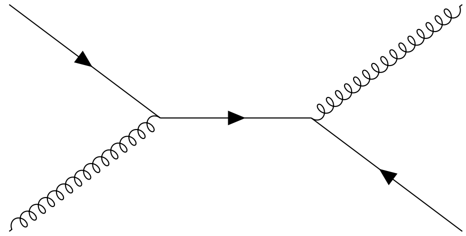


Figure 5.3: $p+p \to gq\,(\bar{q})+X$ events, these dijet-types are possible only via quark (antiquark) and gluon interactions, hence probing both gluon and quark distributions at once

Chapter 6

Jets in Vacuum and the PDF

6.1 Jet Identification Algorithms

As discussed in chapter 1, there are a variety of jet finding algorithms that prioritize different theoretical aspects of the underlying physics while being experimentally realizable [3] [1].

In general, a jet identification algorithm needs to be IRC safe. That is, the jet object needs to display invariance in the Infrared (IR) and Collinear regimes, managing real-virtual cancellation and keeping results meaningful for emission and splitting respectively.

Chapter 7

So exactly how intelligent is AI

Chapter 8

Proof Solving and Validation as Quality Control Mechanisms

Part III

Experimental Output: The Main Event

Chapter 9

So is sPHENIX actually working?

Chapter 10

Measuring the Energy Correlator

10.1 Discretization and Efficiency

10.2 Efficiency corrections

As was discussed in section **??**, the selection of a jet cone limits the accuracy of the energy correlator calculation in a way that depends on R_L. Beyond the problems noted in that model, one also has the issue of normalization and efficienty. Any set of pairs (or tripplets or N-point combinations) measured in a calorimeter needs to be corrected for the overabndence of pairs sitting at the discretized lattice in R_L. To proceed with the efficiency correction on the two point correlator, one must begin with considering the continuous case. Any jet cone limits the number of identified pairs which has the potential to bias the results, thus an efficiency of identification of contianed pairs must be introduced. Begin by drawing a jet cone of radius R_j in the $\eta - \varphi$ plane of some idealized zero-width tower calorimeter, and select some point in the jet offset from the center by some distance and angle λ, then draw another circle of radius R_L around this point limited to the jet cone–representing all pairs to the chosen point that lie within the jet–as show in fig **??**. Then the efficiency ratio defined as geometric pairs in jet over geometric pairs in the phase space gives

$$\begin{aligned}\tilde{\varepsilon}\left(R_L, R_J\right) &= \frac{\int_0^{2\pi} \mathrm{d}\lambda \int_{\max\{0, R_L - R_j\}}^{R_j} \mathrm{d}\text{ꝛ} C_{inc}}{C_{Total}} \\ &= \frac{\int_0^{2\pi} \mathrm{d}\lambda \int_{\max\{0, R_L - R_j\}}^{R_j} \mathrm{d}\text{ꝛ} 2 \int_0^{\theta_f} \mathrm{d}\theta \pi \theta R_L}{\int_0^{2\pi} \mathrm{d}\lambda \int_0^{R_j} \mathrm{d}\text{ꝛ} 2\pi R_L}\end{aligned} \tag{10.1}$$

Where θ is the polar angle measured around the circle of pairs, C_{inc} is the circumfrence of the circles of pairs, C_{Total} is just the total pairs in the phase space and is used as a normalization factor to allow for comparisons with the discretized case. The bounds for the θ integral reflect that two points are symmetric around the axis defined by the centers of the two circles, and $\cos\theta_f = \frac{R_L^2 + \text{ꝛ}^2 - R_j^2}{2\text{ꝛ} R_j^2}$, and the bounds on ꝛ are given to keep at least one pair within the jet in the numerator. Switching the order of integration in eq (10.1) and bounds allows for a version of this integral to be done, and then the resulting elliptical integral is evaluted around $\arccos\left(\frac{R_L}{2R_j}\right)$ for an approximate answer

$$\begin{aligned}\tilde{\varepsilon}\left(R_L, R_j\right) &= \frac{R_L}{2\pi R_j^2}\sqrt{4R_j^2 - R_L^2} + \frac{E\left(\arccos\left(\frac{R_L}{2R_J}\right)\middle|\frac{R_L^2}{R_j^2}\right)}{2\pi} \\ &\approx \sqrt{R_L}2\pi R_j^2\left[\frac{4R_j^2 - R_L^2}{+}R_j\arccos\left(\frac{R_L}{2R_J}\right)\right. \\ &\qquad \left.-\frac{R_L}{3}\left(\arccos\left(\frac{R_L}{2R_J}\right)\right)^3 + \mathcal{O}\left(\left[\arccos\left(\frac{R_L}{2R_J}\right)\right]^5\right)\right]\end{aligned} \tag{10.2}$$

Switching to the discretized case, one introduces the new parameters δ, γ being the size of a calorimeter tower in φ, η respectively. First take $\delta \approx \gamma$ such that the problem is one of square towers, which, for the purposes of analysis of sPHENIX data is appropriate, given that the towers of each calorimeter are almoest square–see sections **??** and **??**. Then, we start with a definition of the circumference to use in this case. One states that all towers can be expressed as a set of points of the form $(p\delta, q\delta)$ s.t. $p, q \in \mathbb{Z}$, and the equivilent measure to the circumference of the continiuous around a central point $P_c = (p_c, q_c)\delta$ can be stated as

$$\left|\mathcal{C}\left[R_L, \delta, (p_c, q_c)\right]\right| = \left|\left\{P \middle| \left(p - p_c\right)^2 + \left(q - q_c\right)^2 = \frac{R_L^2}{\delta^2}\right\}\right| \tag{10.3}$$

A natural extension from this measure tells us that $\frac{R_L^2}{\delta^2} = \mathbb{Z}$ which gives a natural binning for the energy-energy correlator, if plotedd as a function of R_L^2. From here, the equvilent expression to eq. 10.1 can be stated as

$$\tilde{\varepsilon}\left(R_L, R_j, \delta\right) = \frac{\sum_{(m,n)\in \mathrm{Disk}(R_j)} \left|\mathcal{C}\left[R_L, \delta\left(m, n\right)\right] \cap \mathrm{Disk}\left(R_j\right)\right|}{\left|\mathrm{Disk}\left(R_j\right)\right| \left|\mathcal{C}\left[R_L, \delta, (0, 0)\right]\right|} \tag{10.4}$$

where $\mathrm{Disk}(R_j)$ is the collection of points that lie within a circle of radius R_j and $\mathcal{C}$ is the collection of points at exactly R_L from the center point $(m, n)\delta$. The cardinality of these sets can be found by noting that $|\mathcal{C}|$ is simply the sum-of-squares function $r_2\left(\frac{R_L^2}{\delta^2}\right)$ and

$|\text{Disk}\,(R_j)|$ is Gauss's Circle Problem [19].

In order to solve for the numerator, one takes a second disk of radius R'_j such that,

$$\mathcal{C}\left[R_L, \delta\,(m,n) \cap \text{Disk}\left(R'_j\right)\right] = \mathcal{C} \quad \forall (m,n) \in \text{Disk}\,(R_j) \tag{10.5}$$

Which, allows for the folding of the sumation into a problem of set theory, noting that

$$\begin{aligned} \text{Disk}\left(R'_j\right) &= \text{Disk}\,(R_j) \cup \sum_{(m,n)} \mathcal{C} \\ \Rightarrow |\text{Disk}\,(R_j) \cap \sum_{(m,n)} \mathcal{C}| &= |\text{Disk}\,(R_j)| + \left|\sum_{(m,n)} \mathcal{C}\right| - \left|\text{Disk}\left(R'_j\right)\right| \\ &= |\text{Disk}\,(R_j)|\,(1 + |\mathcal{C}\,[R_L, \delta, (0,0)]|) - \left|\text{Disk}\left(R'_j\right)\right| \end{aligned} \tag{10.6}$$

Setting $R'_j = R_j + R_L$ as the largest value for R'_j which corresponds to a collection of pairs centered on one of the edges of the faces, the equivilent to eq 10.2 is simply

$$\tilde{\varepsilon}\,(R_L, R_j, \delta) = 1 - \frac{1}{r_2\left(\frac{R_L^2}{\delta^2}\right)} \left[\sum_{i=\frac{R_j^2}{\delta^2}}^{\frac{\left(R_j+R_L\right)^2}{\delta^2} i} r_2(i) \Bigg/ \sum_{i=0}^{\frac{R_j^2}{\delta^2} i} r_2(i) \right] \tag{10.7}$$

To approximate the solution one can the identity from Hardy and Ramanujan [20], however, for our purposes, it sufficies to use numerical solutions for the special cases that are relevant to the detector as is shown in fig **??** Thus, the full efficiency correction on ΔR_L to correct for both edge effects of the jet cone and the effects of a finite spaitial resolution on the

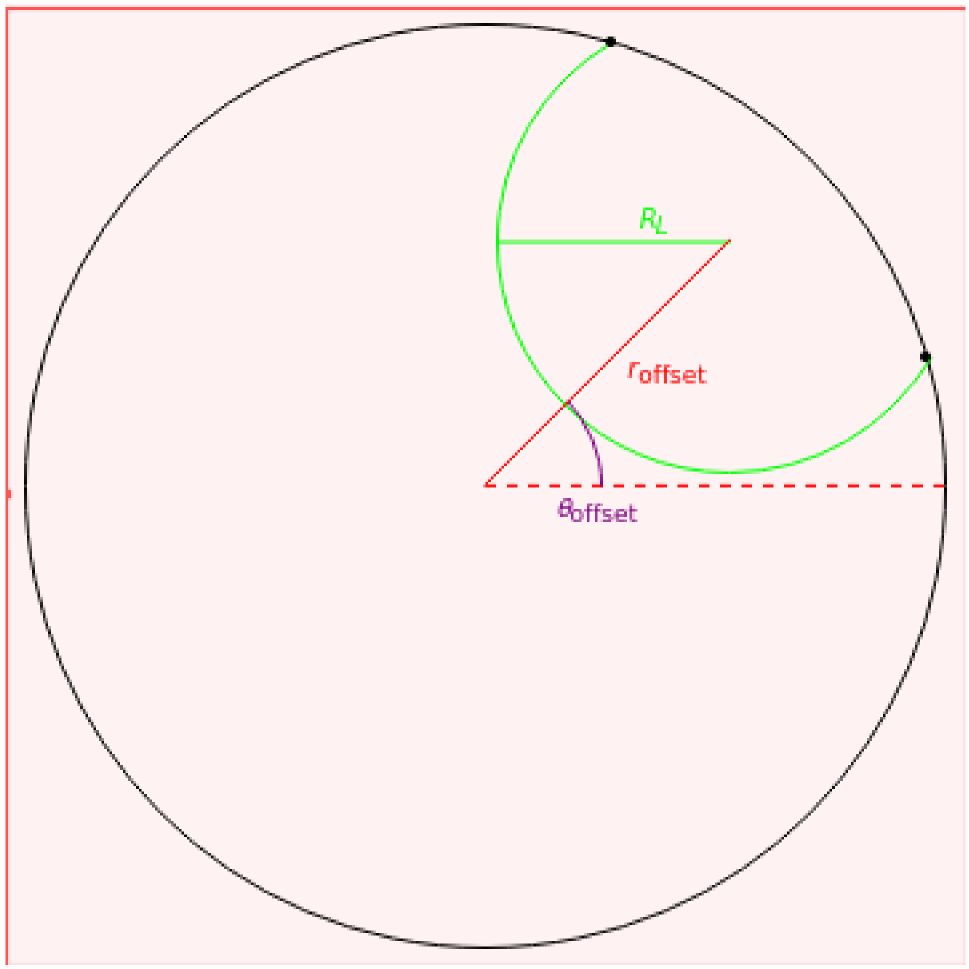


Figure 10.1: The continious and discretized calorimeters for phase space and discretization efficiency correction to smooth distribution and reduce bin width selection dependence

calorimeters is

$$\varepsilon\left(R_L, R_j, \delta\right) = \frac{R_L r_2\left(\frac{R_L^2}{\delta^2}\right)}{2\pi R_j^2}\left[\frac{\sqrt{4R_j^2 - R_L^2} + R_j E\left(\arccos\left(\frac{R_L}{2R_J}\right)\middle|\frac{R_L^2}{R_j^2}\right)}{r_2\left(\frac{R_L^2}{\delta^2}\right) - \sum_{i=\frac{R_j^2}{\delta^2}}^{\frac{\left(R_j+R_L\right)^2}{\delta^2} i} r_2(i) \Big/ \sum_{i=0}^{\frac{R_j^2}{\delta^2} i} r_2(i)}\right] \tag{10.8}$$

10.3 Uncertainties

10.3.1 Statistical Uncertianity

10.3.2 Systematic Uncertianity sources

10.3.3 Binning Uncertianity

10.3.4 Systematic Uncertianities

In the discussion of the energy correlator in chapter 5, we had been working in an idealized regime where the measurement is able to be resolved to arbitary size, however, this is limited by the physical reality of the detector systems.

In simulations and data, one finds that the practical reality of making measurements of the Energy correlator is severly hamstrung by the use of tower size objects rather than the truth particles that make up our idealized toy jets discussed in section 6.1. To study the efficiency loss, one takes the identified truth jet objects from Monte Carlo Simulations (Pythia8 and Herwig for pp; sHIJING, EPOS4, AMPT for AuAu) and runs them through Geant simulations to model detector effects. One then takes the initial truth jet, draws a circle that encloses the truth particles, and projects this out onto the calibrated tower response in each calorimeter, as well as runs an anti-k_{T} jet identification algorithm to simulate jet reconstruction. These three object types (truth jet, projective cone and reconstructed jet) are collected into sets and the measurement of the two- and three-point energy correlators is performed using the defintion in eq 10.9

$$\begin{aligned} \langle E2C\left(R_L\right)\rangle &= \frac{1}{N_{\text{events}}}\sum_{\text{events}}\frac{1}{N_{\text{jet}}}\sum_{N_{\text{jets}}}\sum_{i,j\in\text{Jet Consitutents}}\frac{E_iE_j}{E_{\text{jets}}^2}\delta\left(R_{ij}-R_L\right) \\ \langle E3C\left(R_L\right)\rangle &= \frac{1}{N_{\text{events}}}\sum_{\text{events}}\frac{1}{N_{\text{jet}}}\sum_{N_{\text{jets}}}\sum_{i,j,k\in\text{Jet Consitutents}}\frac{E_iE_jE_k}{E_{\text{jets}}^3}\delta\left(\max R_{i,j,k}-R_L\right) \end{aligned} \tag{10.9}$$

10.4 Whole Calorimeter Measurements

While the Energy correlator is generally studied within a jet, one can make use of the whole calorimeter to compare the energy correlator across all angular seperations to characterize the underlying event. This induces a new computational issue of scale, where the number of calculations goes from ${}_{50}C_2 = 1225$ in the HCAL and ${}_{804}C_2 = 322,806$ in the EMCAL to ${}_{1536}C_2 \approx 2.4\times10^6$ and ${}_{24576}C_2 \approx 6\times10^8$ in the EMCAL. However, since pp events

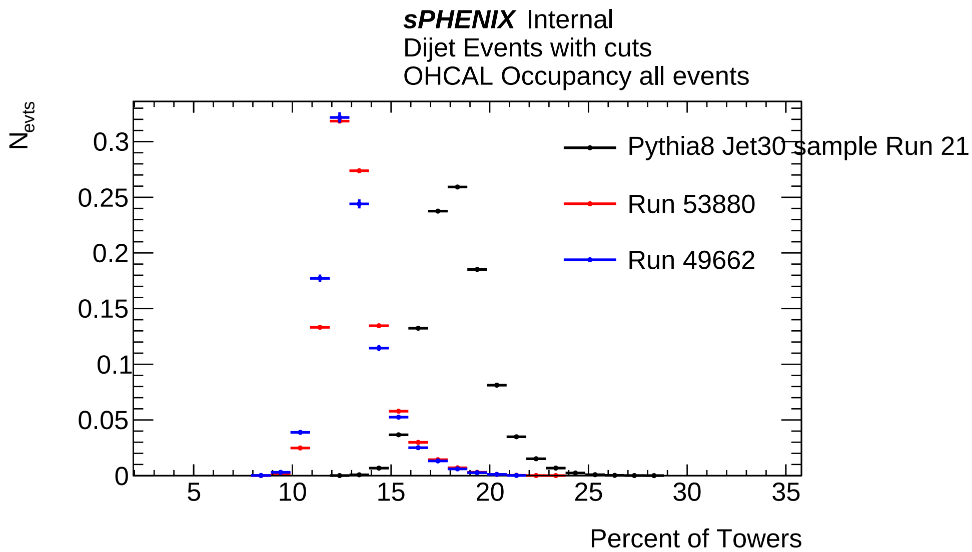


Figure 10.2: Multiplicty in each Calorimeter (pending–have ohcal now) as percentage of towers with $E \geq 10$ MeV to filter out electronics fluctuations

have relatively low multiplicity in the calormeters as is seen in fig 10.2, one reduces the computational load signifigantly by preprocessing the towers to remove all towers with energy below some threshold, functionally mirroring the zero-suppression procedure discussed in section 2.1.1.

One also notes that the whole event correlator on calorimeter towers is signfigantly sensitive to the electronic noise of the towers, given that any positive electronic noise would mimic a valid energy deposition. While the zero supression reduces the noise by a signifigant fraction at the hardware level, there are fluctations that pass this cut. Each event has an entry for every tower, and due to this electronic noise not having the same waveform shape in the ADC (see figs **??** and **??** for a comparision of a pedestal waveform and that of a data event–there are a fraction of towers that have an apparent negative energy, which when summing the total energy of the calorimeters, approximately cancels out positive fluctionations on average. However in our case, it is not so simple. Consider the following ansatz of a symmetric distribution $S = \{-2, -2, -1, -1, -1, -1, 1, 1, 1, 1, 1, 2, 2\}$, then one considers the sum of the product of pairs, being the numerator of the two point energy correlator would be the only way to have the noise contribution go to zero

$$
\begin{aligned}
PS &= (-2)^2 + (-2 \cdot -1) \times 8 + (-2 \cdot 1) \times 8 + (-2 \cdot 2) \times 2 + (-1)^2 \times 3! \\
&\quad + (-1 \cdot 1) \times 16 + (-1 \cdot 2) \times 8 + (1)^2 \times 3! + (1 \cdot 2) \times 8 + 2^2 \\
&= -4
\end{aligned} \tag{10.10}
$$

Clearly a symmetric distribution alone does not garuntee a full cancellation. For a full proof of this for a gaussian distribution see appendix A.

Chapter 11

The Power of the ENC: α_S at the few GeV scale

As had been shown recently at CMS [21], taking the ratio of the two and (projective) three point energy correlator results in a distribution that scales directly with $\alpha_s(Q^2)R_L$ where Q^2 is the scale of the process under consideration. To make this calculation in the measurement presented in this thesis, we need to fix the hard scale relevant to the event. Here it is instructive to consider how we can adapt the methods utilized for a jet based energy correlator measurment to those for an event based model. In the CMS analysis is [21], they use a definition of the perturbative scale and hard scale as

$$\begin{aligned} PS &= \frac{p_T^{jet} R}{2} \\ Q^2 &= \left(p_1^{jet} + p_2^{jet}\right)^2 \end{aligned} \tag{11.1}$$

Chapter 12

Event Classification

The more standard method of measuring α_S requires that one be able to accurately determine the relative quark and gluon jet populations in the event samples [22]. While the discussion in chapter 11 allow for us to circumvent this issue in the immediate measurement of the strong force coulpling, it is still a useful quantity to measure for further studies on jet substructure measurements. So while it is inapproriate to tag individual jets or events as a "quark" or "gluon" type event through the Energy Correlator, given that any individual jet may have fluctuations that overlap into

Chapter 13

Comparison is the Theft of Joy: What does the LHC say, and how about STAR?

Chapter 14

Entanglement and other Lofty Goals

In this thesis, I have presented a limited scope measurement of a whole event energy correlator in p+p event which still empowers us to use it to measure α_S and to estimate our gluon and quark jet populations.

Part IV

Wrapping it all up

Chapter 15

Observable prospects for p+p in run 24, AuAu with run 24 and 25 and the EIC

Chapter 16

Remaining Questions

Chapter 17

Implementation and application for the remaining sPHENIX data

Part V

Odds and Ends-Matter

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Appendices

Appendix A

Proof that a Symmetric Energy Distribution fails to cancel in the Product-Sum

In the ansatz reflected in equation (10.10), a symmetric set with a distribution roughly approximating a gaussian was utilized, however, we can go beyond that to properly show this result is the case in a set drawn from a generalized Gaussian distribution

Proof

We begin with a gaussian distribution $P(x)$ and sample points from it such that we construct a "Gaussian sample set" with the following properties

Definition A.0.1 (Gaussian Sample Set)**.** Take a set S that is constructed of points randomly sampled such that

$$S \approx P(x) = e^{\frac{-x^2}{\sigma}} \tag{A.1}$$

Which gives the following properties

1. $S = \{x|\bar{x} = \delta_{\bar{x}}\}$
2. $\frac{1}{N}\sum_{x\in S} x^2 = (\sigma + \delta_\sigma)^2$
3. $|\delta_{\bar{x}}|, |\delta_\sigma| << 1$
4. $\lim_{N\to\inf} |\delta_{\bar{x}}|, |\delta_\sigma| = 0$

Then, the Product-Sum is calculated by following the standard algorithm

Algorithm: PRODUCT-SUM

$$\begin{aligned} &\forall x \in S s.t. len(S) > 1 \\ &\quad S = S/x \\ &\quad PS = PS + (x * y) \qquad\qquad \forall y \in S \end{aligned} \tag{A.2}$$

Product-sum algorithm for the sample set S. Initially the PS varaiable is set at 0

Using the properties of S noted in definition A.0.1, one notes that for any sampled value x, there should be approximately $n = \left\lfloor \frac{1}{\sqrt{2\pi(\sigma+\delta_\sigma}} e^{\frac{(x-\delta_{\bar{x}})^2}{\sigma+\delta_\sigma}} \right\rfloor$ of the same value. In fact, by including these small corrections and an additional small correction $|\delta_{n,x}| << 1$ we fix this number to the Gaussian curve within some fluctations. With this in mind, one breaks the contributions to the Product-sum into two categories, the $y = x$ and $y \neq x$ case. For the first case, the contribution is simply

$$PS_{y=x} = x^2 \left(n(x) - 1\right)! \tag{A.3}$$

and in the second case

$$PS_{y\neq x} = xyn(x)n(y) \tag{A.4}$$

Thus the product-sum becomes

$$PS = \sum_{x \in S} \left[x^2 \left(n(x) - 1\right)! + \sum_{y > x, y \in S} xyn(x)n(y) \right] \tag{A.5}$$

With a sufficiently high number of samples, we can shift to the continuum limit and write

$$\tilde{PS} = \int_{-\infty}^{\infty} \mathrm{d}x \ \left[x^2 \Gamma((n(x) - 1)) + \int_{x}^{\infty} \mathrm{d}y \frac{xy}{2\pi\left(\sigma + \delta\sigma\right)} e^{\frac{-(x-\delta_{\bar{x}})^2 - (y-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \right] \tag{A.6}$$

$\mathcal{QED}$